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| APPLICATION NO. | FILING DATE | FIRST NAMED INVENTOR | ATTORNEY DOCKET NO. | CONFIRMATION NO. |
|---|---|---|---|---|
| 18/698,915 | 04/05/2024 | Tetsu MORISHIMA | 228963 | 3163 |

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| EXAMINER | |
|---|---|
| WONG, ERIC K | |

| ART UNIT | PAPER NUMBER |
|---|---|
| 2874 | |

| NOTIFICATION DATE | DELIVERY MODE |
|---|---|
| 06/17/2026 | ELECTRONIC |

**Please find below and/or attached an Office communication concerning this application or proceeding.**

The time period for reply, if any, is set in the attached communication.

Notice of the Office communication was sent electronically on above-indicated "Notification Date" to the following e-mail address(es):

OfficeAction25944@oliff.com
jarmstrong@oliff.com

| Office Action Summary | Application No. 18/698,915 | Applicant(s) MORISHIMA et al. | |
|---|---|---|---|
| | Examiner Eric K Wong | Art Unit 2874 | AIA (First Inventor to File) Status Yes |

*-- The MAILING DATE of this communication appears on the cover sheet with the correspondence address --*

**Period for Reply**

A SHORTENED STATUTORY PERIOD FOR REPLY IS SET TO EXPIRE 3 MONTHS FROM THE MAILING DATE OF THIS COMMUNICATION.

- Extensions of time may be available under the provisions of 37 CFR 1.136(a). In no event, however, may a reply be timely filed after SIX (6) MONTHS from the mailing date of this communication.
- If NO period for reply is specified above, the maximum statutory period will apply and will expire SIX (6) MONTHS from the mailing date of this communication.
- Failure to reply within the set or extended period for reply will, by statute, cause the application to become ABANDONED (35 U.S.C. § 133). Any reply received by the Office later than three months after the mailing date of this communication, even if timely filed, may reduce any earned patent term adjustment. See 37 CFR 1.704(b).

**Status**

1)☑ Responsive to communication(s) filed on 4/14/2026.

☐ A declaration(s)/affidavit(s) under **37 CFR 1.130(b)** was/were filed on _____.

2a)☑ This action is **FINAL.** 2b) ☐ This action is non-final.

3)☐ An election was made by the applicant in response to a restriction requirement set forth during the interview on _____; the restriction requirement and election have been incorporated into this action.

4)☐ Since this application is in condition for allowance except for formal matters, prosecution as to the merits is closed in accordance with the practice under *Ex parte Quayle*, 1935 C.D. 11, 453 O.G. 213.

**Disposition of Claims***

5) ☑ Claim(s) 1-8 is/are pending in the application.

5a) Of the above claim(s) _____ is/are withdrawn from consideration.

6) ☐ Claim(s) _____ is/are allowed.

7) ☑ Claim(s) 1-8 is/are rejected.

8) ☐ Claim(s) _____ is/are objected to.

9) ☐ Claim(s) _____ are subject to restriction and/or election requirement

* If any claims have been determined allowable, you may be eligible to benefit from the **Patent Prosecution Highway** program at a participating intellectual property office for the corresponding application. For more information, please see http://www.uspto.gov/patents/init_events/pph/index.jsp or send an inquiry to **PPHfeedback@uspto.gov.**

**Application Papers**

10)☐ The specification is objected to by the Examiner.

11)☐ The drawing(s) filed on _____ is/are: a)☐ accepted or b)☐ objected to by the Examiner.

Applicant may not request that any objection to the drawing(s) be held in abeyance. See 37 CFR 1.85(a).

Replacement drawing sheet(s) including the correction is required if the drawing(s) is objected to. See 37 CFR 1.121(d).

**Priority under 35 U.S.C. § 119**

12)☐ Acknowledgment is made of a claim for foreign priority under 35 U.S.C. § 119(a)-(d) or (f).

**Certified copies:**

a)☐ All b)☐ Some** c)☐ None of the:

1.☐ Certified copies of the priority documents have been received.

2.☐ Certified copies of the priority documents have been received in Application No. _____.

3.☐ Copies of the certified copies of the priority documents have been received in this National Stage application from the International Bureau (PCT Rule 17.2(a)).

** See the attached detailed Office action for a list of the certified copies not received.

**Attachment(s)**

1) ☑ Notice of References Cited (PTO-892)

2) ☑ Information Disclosure Statement(s) (PTO/SB/08a and/or PTO/SB/08b) Paper No(s)/Mail Date _____.

3) ☐ Interview Summary (PTO-413) Paper No(s)/Mail Date _____.

4) ☐ Other: _____.

## DETAILED ACTION

### *Response to Arguments*

1. Applicant's arguments with associated claim amendments, filed 4/14/2026, have been fully considered and are persuasive. Therefore, the prior rejections have been withdrawn. However, upon further consideration, a new ground(s) of rejection is made in view of US 2020/0166715 to Shimakawa et al.

2. Applicant's arguments relate to newly clarified limitations regarding the reference surfaces and the planar surfaces that are parallel with a flange. These elements appear to be disclosed by Shimikawa in figures 5-6 and 11.

### *Claim Rejections - 35 USC § 102*

3. The following is a quotation of the appropriate paragraphs of 35 U.S.C. 102 that form the basis for the rejections under this section made in this Office action:

> A person shall be entitled to a patent unless –
>
> (a)(1) the claimed invention was patented, described in a printed publication, or in public use, on sale, or otherwise available to the public before the effective filing date of the claimed invention.

4. Claim(s) 1-2 and 4-7 is/are rejected under 35 U.S.C. 102(a)(1) as being anticipated by Shimikawa.

Shimikawa discloses in the abstract and figures, an optical connector comprising:

- a glass fiber (F) and a resin coating covering the glass (film F2),
- a front end portion (towards F1 in figure 8);
- a rear end portion (end towards 7 in figure 8) located on a side opposite to the front end portion (figure 8); and

- a ferrule assembly (2) having a ferrule fixed to a tip end portion (2a) and a holding portion (23; figure 9) to which a rear end portion of the ferrule is fixed and which is provided with a flange (3);
- a housing having an inner wall surface that defines an internal space in which at least the holding portion of the ferrule assembly is accommodated, a flat surface that constitutes a part of the inner wall surface and is provided at a position facing a part of an outer peripheral surface of the flange, and a positioning portion that limits movement of the holding portion in the internal space (figures 11-12 show how the holding portion secures the ferrule in the housing); and
- an elastic member (5) that biases the flange toward the positioning portion,
- wherein the reference surface includes two contact parts (surfaces to the left and right of 23k in figure 11) provided apart from each other by a predetermined distance (the area bounded by spring 23k) and restricting a rotational direction of a flange as a result of the planar surface of the flange being brought into parallel contact with the planar surface of the inner wall surface with the planar surface of the flange contacting the two contact parts (when the flange is assembled with the provided spring cutout portion that resides within a channel that creates two contact points, the flange cannot rotate).

As to claim 2, the reference surface has a recessed part (23k) between the two flat reference parts.

As to claim 4, the claim recites a general "floating" terminology defined as the position changes as the ferrule is pushed through. Figure 8 shows some degree of movement and the spring allows for movement in the longitudinal direction.

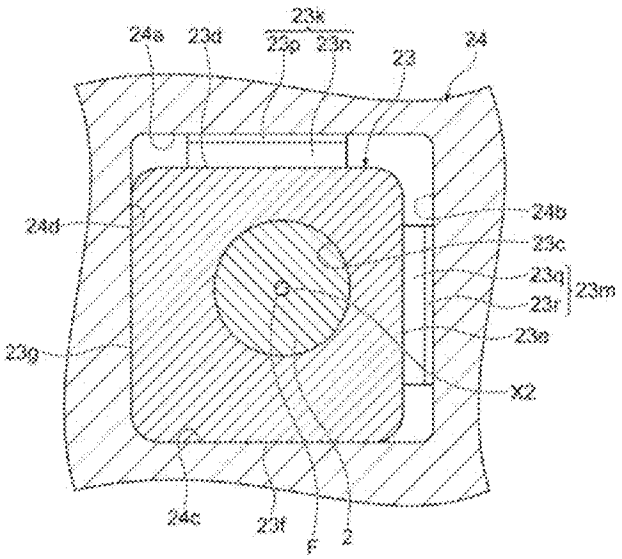
As to claim 5, the outer shape of the flange is quadrangular (figure 11).

As to claim 6, an outer portion is curved (unlabeled corners are curved in figure 11).

As to claim 7, a multicore fiber is disclosed (paragraph 87).

As to claim 8, the "line" extending along the reference surface coincides with the flange (figure 11).

*Fig.11*



***Claim Rejections - 35 USC § 103***

5. In the event the determination of the status of the application as subject to AIA 35 U.S.C. 102 and 103 (or as subject to pre-AIA 35 U.S.C. 102 and 103) is incorrect, any correction of the statutory basis (i.e., changing from AIA to pre-AIA) for the rejection will not be considered a new ground of rejection if the prior art relied upon, and the rationale supporting the rejection, would be the same under either status.

6. The following is a quotation of 35 U.S.C. 103 which forms the basis for all obviousness rejections set forth in this Office action:

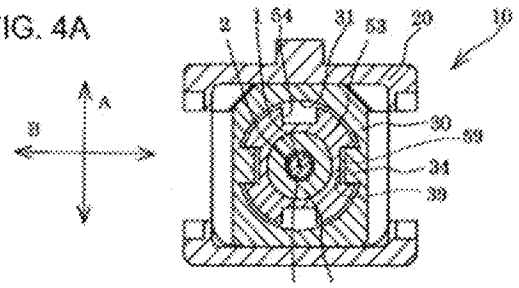
> A patent for a claimed invention may not be obtained, notwithstanding that the claimed invention is not identically disclosed as set forth in section 102, if the differences between the claimed invention and the prior art are such that the claimed invention as a whole would have been obvious before the effective filing date of the claimed invention to a person having ordinary skill in the art to which the claimed invention pertains. Patentability shall not be negated by the manner in which the invention was made.

7. Claim(s) 3 is/are rejected under 35 U.S.C. 103 as being unpatentable over Shimikawa in view of Inagaki.

Shimikawa fails to disclose the recessed reference surface with corresponding projection. Shimikawa does disclose two different spring portions (23p and 23n) which project outwardly when viewed from a front view such as in figure 10. For purposes of examination and to expedite prosecution, Examiner will interpret the projection to mean more than this mere stepped area of the spring in the above view with a notch added. It is noted that such alignment structures are common in the art.

Inagaki discloses such a well known recessed surface with projections (figure 4A).

FIG. 4A



It would have been obvious to one having ordinary skill in the art to add recessed portions with corresponding projections as taught by Inagaki to lock a flange's spring in Shimikawa in place to prevent misalignment.

***Conclusion***

8. Applicant's amendment necessitated the new ground(s) of rejection presented in this Office action. Accordingly, **THIS ACTION IS MADE FINAL**. See MPEP § 706.07(a). Applicant is reminded of the extension of time policy as set forth in 37 CFR 1.136(a).

A shortened statutory period for reply to this final action is set to expire THREE MONTHS from the mailing date of this action. In the event a first reply is filed within TWO MONTHS of the mailing date of this final action and the advisory action is not mailed until after the end of the THREE-MONTH shortened statutory period, then the shortened statutory period will expire on the date the advisory action is mailed, and any nonprovisional extension fee (37 CFR 1.17(a)) pursuant to 37 CFR 1.136(a) will be calculated from the mailing date of the advisory action. In no event, however, will the statutory period for reply expire later than SIX MONTHS from the mailing date of this final action.

9. Any inquiry concerning this communication or earlier communications from the examiner should be directed to Eric K Wong whose telephone number is (571)272-2363. The examiner can normally be reached M-Tu, Th-F 8A-6P.

Examiner interviews are available via telephone, in-person, and video conferencing using a USPTO supplied web-based collaboration tool. To schedule an interview, applicant is encouraged to use the USPTO Automated Interview Request (AIR) at http://www.uspto.gov/interviewpractice.

If attempts to reach the examiner by telephone are unsuccessful, the examiner's supervisor, Thomas Hollweg can be reached at 571-270-1739. The fax phone number for the organization where this application or proceeding is assigned is 571-273-8300.

Information regarding the status of published or unpublished applications may be obtained from Patent Center. Unpublished application information in Patent Center is available to registered users. To file and manage patent submissions in Patent Center, visit: https://patentcenter.uspto.gov. Visit https://www.uspto.gov/patents/apply/patent-center for more information about Patent Center and https://www.uspto.gov/patents/docx for information about filing in DOCX format. For additional questions, contact the Electronic Business Center (EBC) at 866-217-9197 (toll-free). If you would like assistance from a USPTO Customer Service Representative, call 800-786-9199 (IN USA OR CANADA) or 571-272-1000.

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